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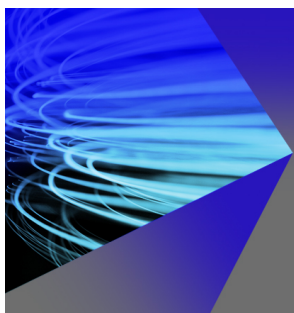
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Inverse problems for the Schrödinger equations with time-dependent electromagnetic potentials and the Aharonov–Bohm effect

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We consider the inverse boundary value problem for the Schrödinger operator with time-dependent electromagnetic potentials in domains with obstacles. We extend the results of the author's works [Inverse Probl. **19**, 49 (2003); **19**, 985 (2003); **20**, 1497 (2004)] to the case of time-dependent potentials. We relate our results to the Aharonov–Bohm effect caused by magnetic and electric fluxes. © 2008 American Institute of Physics. [DOI: [10.1063/1.2841329](https://doi.org/10.1063/1.2841329)]

I. INTRODUCTION

Let Ω_0 be a smooth simply connected bounded domain in $\mathbf{R}^n$. Let $D \subset \Omega_0 \times [0, T]$ be a domain with the following properties.

Denote $D_{t_0} = D \cap \{t = t_0\}$. We assume that $D_{t_0} = \Omega_0 \setminus \bigcup_{j=1}^m \overline{\Omega_j(t_0)}$ where $\Omega_j(t_0)$ are piecewise smooth nonintersecting domains, $1 \leq j \leq m$, $0 \leq t_0 \leq T$. We assume that $\Omega'(t_0) = \bigcup_{j=1}^m \Omega_j(t_0)$ depends smoothly on $t_0 \in [0, T]$.

Finally we assume that the normal to $\partial D \setminus (D_0 \cup D_T)$ in $\mathbf{R}^{n+1}$ is not parallel to the t -axis for any $t \in [0, T]$.

Consider the Schrödinger equation with time-dependent electromagnetic potentials in D as follows:

$$i \frac{\partial u(x, t)}{\partial t} = H u = \sum_{j=1}^n \left(-i \frac{\partial}{\partial x_j} - A_j(x, t) \right)^2 u(x, t) + V(x, t) u(x, t), \quad (x, t) \in D, \quad (1.1)$$

with zero initial conditions

$$u(x, 0) = 0, \quad (x, 0) \in D_0, \quad (1.2)$$

and the Dirichlet boundary conditions

$$u|_{\partial \Omega_0 \times (0, T)} = f, \quad (1.3)$$

We assume that $A(x, t)$, $V(x, t)$ are smooth in $\overline{D}$. Without loss of generality we can assume that $A(x, t) = 0$, $V(x, t) = 0$ near $\partial \Omega_0 \times [0, T]$ (see Remark 3.1). Let Λ be the Dirichlet-to-Neumann (D-to-N) operator on $\partial \Omega_0 \times [0, T]$, i.e.,

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$$\Lambda f = \frac{\partial u}{\partial \nu} - i(A \cdot \nu) u|_{\partial\Omega_0 \times (0,T)}, \quad (1.4)$$

where f is the same as in (1.3) and ν is the outward unit normal to ∂D_t . Denote by $G(D)$ the group of $C^\infty(\bar{D})$ functions $c(x,t)$ such that $c(x,t) \neq 0$ in $\bar{D}$ and denote by $G_0(D)$ the subgroup of $G(D)$ such that $c(x,t) = 1$ on $\partial\Omega_0 \times [0,T]$. We say that the electromagnetic potentials $(A(x,t), V(x,t))$ and $(A'(x,t), V'(x,t))$ are gauge equivalent if there exists $c(x,t) \in G(D)$ such that

$$\begin{aligned} A'_j(x,t) &= A_j(x,t) + ic^{-1}(x,t) \frac{\partial c}{\partial x_j}, \quad 1 \leq j \leq n, \\ V'(x,t) &= V(x,t) - ic^{-1}(x,t) \frac{\partial c}{\partial t}. \end{aligned} \quad (1.5)$$

Note that if $(i(\partial/\partial t) - H)u = 0$ and

$$u' = c^{-1}u, \quad (1.6)$$

then $(i(\partial/\partial t) - H')u' = 0$, where H' is the operator with potentials $A'(x,t), V'(x,t)$. The group $G(D)$ is called the gauge group and (1.6) is called the gauge transformation. When $A(x,t)$ and $V(x,t)$ are real valued it is natural to consider only $c(x,t)$ such that $|c(x,t)| = 1$. Also when D is simply connected any $c(x,t) \in G(D)$ has a form $c(x,t) = e^{i\varphi(x,t)}$, where $\varphi(x,t) \in C^\infty(\bar{D})$.

We say that the D-to-N operators Λ and Λ' corresponding to the Schrödinger operators $i(\partial/\partial t) - H$ and $i(\partial/\partial t) - H'$ are gauge equivalent if there exists $c(x,t) \in G(D)$ such that

$$\Lambda' = c_0^{-1} \Lambda c_0 \quad \text{on } \partial\Omega_0 \times [0,T], \quad (1.7)$$

where c_0 is the restriction of $c(x,t)$ to $\partial\Omega_0 \times [0,T]$.

We shall introduce gauge invariant boundary data on $\partial\Omega_0 \times (0,T)$ (cf. Ref. 4). Let $u(x,t)$ be a solution of $i(\partial/\partial t) - Hu = 0$ in D , $u(x,0) = 0$.

Let $|u(x,t)|^2$ be the probability density and

$$S(x,t) = \Im \left(\frac{\partial u(x,t)}{\partial x} - iA(x,t)u(x,t) \right) \overline{u(x,t)}$$

be the probability current. We define the boundary data of $u(x,t)$ on $\partial\Omega_0 \times (0,T)$ as

$$|u(x,t)|^2|_{\partial\Omega_0 \times (0,T)} = f_1, \quad \frac{\partial}{\partial \nu} |u(x,t)|^2|_{\partial\Omega_0 \times (0,T)} = f_2, \quad S(x,t)|_{\partial\Omega_0 \times (0,T)} = f_3, \quad (1.8)$$

where $\partial/\partial \nu$ is the normal derivative [cf. (1.4)]. Note that the probability density and the probability current are gauge invariant.

The following proposition was proven in Ref. 4.

Proposition 1.1: *Let $(i(\partial/\partial t) - H)u = 0$ and $(i(\partial/\partial t) - H')u' = 0$ be two Schrödinger equations in D , $u(x,0) = u'(x,0) = 0$, $u(x,t)|_{\partial\Omega'(t)} = u'(x,t)|_{\partial\Omega'(t)} = 0$ for all $t \in [0,T]$. Then the D-to-N operators Λ and Λ' are gauge equivalent on $\partial\Omega_0 \times (0,T)$ if and only if for any solution u there exists u' such that the gauge invariant boundary data (1.8) of $u(x,t)$ and $u'(x,t)$ are equal.*

This paper is a completion of works⁴⁻⁶ on the inverse boundary value problems for the stationary Schrödinger equation in a situation where the Aharonov–Bohm effect caused by magnetic fluxes holds. Here we extend these results to the time-dependent Schrödinger equation with time-dependent electromagnetic potentials. In this case the Aharonov–Bohm effect is caused by both the magnetic and electric fluxes.

Let $i(\partial/\partial t) - H^{(p)}$, $p = 1, 2$ be two Schrödinger operators with time-dependent electromagnetic potentials $(A^{(p)}(x,t), V^{(p)}(x,t))$, $p = 1, 2$, in the domain D described above.

Theorem 1.1: Assume that $n \geq 2$ and the domain D is such that for any $t_0 \in [0, T]$, D_{t_0} satisfies the following condition (cf. Ref. 5, p. 287): All $\Omega_j(t_0)$ are convex, $1 \leq j \leq m$, and for each point $x_0 \in D_{t_0}$ there exists a two-dimensional plane $\Pi \subset \mathbb{R}^n$ containing x_0 such that Π intersects at most one domain $\Omega_j(t_0)$, $1 \leq j \leq m$. Let $i(\partial/\partial t) - H_j$, $j=1, 2$ be two Schrödinger operators in D and $\Lambda^{(j)}$, $j=1, 2$ be the corresponding D -to- N operator on $\partial\Omega_0 \times (0, T)$. If $\Lambda^{(1)}$ and $\Lambda^{(2)}$ are gauge equivalent on $\partial\Omega_0 \times (0, T)$ then the electromagnetic potentials $A^{(1)}(x, t), V^{(1)}(x, t)$ and $A^{(2)}(x, t), V^{(2)}(x, t)$ are gauge equivalent in D .

Theorem 1.1 generalized the result of Ref. 5. Note that when $n=2$ the domain D_{t_0} contains at most one convex $\Omega_j(t_0)$.

The following theorem generalizing the result of Ref. 6 treats the case when D_{t_0} contains more than one obstacle $\Omega_j(t_0)$ in the two-dimensional case.

Theorem 1.2: Assume that $n=2$ and that for each $t_0 \in [0, T]$ domains $\Omega_j(t_0)$, $1 \leq j \leq m$ are smooth except a finite number of corner points and are convex. Suppose also that there is no trapping broken rays [cf. conditions (a) and (b) in Ref. 6, p. 1507]: any broken ray in D_{t_0} , starting at $\partial\Omega_0$, avoiding corner points and having only nontangential reflections at $\partial\Omega'(t_0)$, ends at $\partial\Omega_0$ after a finite number of reflections. The number of reflections is assumed to be uniformly bounded for all such broken rays. If D -to- N operators $\Lambda^{(1)}$ and $\Lambda^{(2)}$ are gauge equivalent on $\partial\Omega_0 \times (0, T_0)$, then the electromagnetic potentials $A^{(1)}(x, t), V^{(1)}(x, t)$ and $A^{(2)}(x, t), V^{(2)}(x, t)$ are gauge equivalent on D .

Inverse problems and Aharonov–Bohm effect were considered in Refs. 14 and 21, ^{4,6} (see also a recent paper³). General inverse problems for the stationary Schrödinger equations can be reduced to the inverse problems for the hyperbolic equations with the time-independent coefficients (see Refs. 2, 13, 9, and 10, and additional references there). Hyperbolic equation approach is very powerful. However, it does not apply to the Schrödinger equations with time-dependent potentials.

Note that the inverse problem for the Schrödinger equation with time-periodic potential was considered earlier in Ref. 22.

The plan of the paper is the following: In Sec. II we prove Theorems 1.1 and 1.2. In Sec. III we consider the Schrödinger operators with time-dependent Yang–Mills potentials. In Sec. IV we discuss the Aharonov–Bohm effect (see Refs. 1, 23, and 17) in the situations covered by Theorems 1.1 and 1.2. Note that the Aharonov–Bohm effect is caused by the magnetic and electric fluxes, and the case of electric fluxes appears only when potentials are time dependent.

II. THE PROOF OF THEOREMS 1.1 AND 1.2

We shall start with the construction of geometric optics solutions of (1.1). We are looking for a solution in the form

$$u_{N0} = e^{-ik^2 t + ik(x \cdot \omega)} a_0^{(N)}, \quad (2.1)$$

where $a_0^{(N)} = \sum_{p=0}^N a_{p0}(x, t, \omega) / (ik)^p$, $|\omega|=1$, k is a large parameter. Substituting u_{N0} in (1.1) we obtain

$$k^2 a_0^{(N)} + i \frac{\partial a_0^{(N)}}{\partial t} = \left(-i \frac{\partial}{\partial x} + k\omega - A(x, t) \right)^2 a_0^{(N)} + V(x, t) a_0^{(N)}.$$

Equating the equal power of k we get

$$\omega \cdot \left(-i \frac{\partial}{\partial x} - A \right) a_{00} = 0, \quad (2.2)$$

$$2\omega \cdot \left(-i \frac{\partial}{\partial x} - A \right) a_{p0} = \left(i \frac{\partial}{\partial t} - H \right) a_{p-1,0}, \quad p \geq 1. \quad (2.3)$$

To solve (2.2) denote $s = \omega \cdot x$, $\tau_j = \omega_{\perp j} \cdot x$, $1 \leq j \leq n-1$, where $\omega_{\perp j}$, $1 \leq j \leq n-1$ is an orthonormal basis in the orthogonal complement to ω in $\mathbf{R}^n$. We have $\partial a_{00} / \partial s - i(A \cdot \omega) a_{00}(s, \tau, t) = 0$, where $\tau = (\tau_1, \dots, \tau_{n-1})$. Therefore we can take

$$a_{00}(s, \tau, t) = \chi_1(t) \chi_2(\tau) \exp \left(i \int_{s_0}^s A \left(\sum_{j=1}^{n-1} \tau_j \omega_{\perp j} + s' \omega, t \right) \cdot \omega ds' \right), \quad (2.4)$$

where $\chi_1(t) = (1/\sqrt{\varepsilon}) \chi_0((t-t_0)/\varepsilon)$, $\chi_2(\tau) = \frac{1}{\varepsilon^{(n-1)/2}} \prod_{j=1}^{n-1} \chi_0((\tau_j - \tau_{0j})/\varepsilon)$, $\chi_0 \in C_0^\infty(\mathbf{R}^1)$, $\chi_0(t) = 0$ for $|t| > 1$, $\int_{-\infty}^\infty \chi_0^2(t) dt = 1$.

We solve (2.3) prescribing the initial condition

$$a_{p0}(s_0, \tau, t) = 0, \quad 1 \leq p \leq N. \quad (2.5)$$

Here s_0, τ_0, t_0 are such that the point $(x_0^{(0)} = s_0 \omega + \sum_{j=1}^{n-1} \tau_{0j} \omega_{\perp j}, t_0)$ does not belong to $\Omega_0 \times [0, T]$ and the plane $(x - x_0^{(0)}) \cdot \omega = 0$ does not intersect Ω_0 . We have

$$\left(i \frac{\partial}{\partial t} - H \right) u_{N0} = \frac{e^{-ik^2 t + ik(x \cdot \omega)}}{(ik)^N} f_{N0},$$

where $f_{N0} = (i(\partial/\partial t) - H) a_{N0}$.

Until now we assumed that the ray $x = (s - s_0)\omega + x_0^{(0)}$ does not meet the obstacles. In the case when this ray meets an obstacle we have to consider a broken ray that reflects at $\partial\Omega_j(t_0)$, $1 \leq j \leq m$ (cf. Ref. 6, pp. 1498–1499). More precisely, we say that $\gamma = \gamma_0 \cup \gamma_1 \cup \dots \cup \gamma_r$ is a broken ray with legs $\gamma_0, \gamma_1, \dots, \gamma_r$ if γ_0 is the ray $x = x_0^{(0)} + (s - s_0)\omega$, $s_0 \leq s \leq s_1$, γ_0 hits $\partial\Omega'(t_0) = \bigcup_{j=1}^m \partial\Omega_j(t_0)$ at some point $x_0^{(1)} = x_0^{(0)} + (s_1 - s_0)\omega \in \partial\Omega'(t_0)$, γ_1 is the reflected ray. We assume that $x_0^{(1)}$ is not a tangential point of reflection. Then

$$\theta_1 = \theta_0 - 2(n(x_0^{(1)}, t_0) \cdot \theta_0) n(x_0^{(1)}, t_0), \quad (2.6)$$

where $\theta_0 = \omega$ is the direction of γ_0 , θ_1 is the direction of γ_1 , and $n(x_0^{(1)}, t_0)$ is the outward unit normal to $\partial\Omega'(t_0)$ at the point $x_0^{(1)}$. Analogously, $\gamma_2, \dots, \gamma_{r-1}$ have nontangential points of reflection at $\partial\Omega'(t_0)$ and γ_r ends at $\partial\Omega_0 \times \{t = t_0\}$.

We associate with broken ray γ the following geometric optics solution (cf. Ref. 6, p. 1499):

$$u_N = \sum_{j=0}^r \sum_{p=0}^N \frac{a_{pj}(x, t, \omega)}{(ik)^p} e^{-ik^2 t + ik\psi_j(x, t, \omega)}, \quad (2.7)$$

where $\psi_0(x, t, \omega) = x \cdot \omega$, $a_{p0}(x, t, \omega)$ are the same as in (2.2)–(2.5); eiconals $\psi_j(x, t, \omega)$ satisfy the equations

$$\left| \frac{\partial \psi_j(x, t, \omega)}{\partial x} \right| = 1, \quad (2.8)$$

$$\psi_j(x, t, \omega)|_{\partial\Omega'(t)} = \psi_{j+1}(x, t, \omega)|_{\partial\Omega'(t)}, \quad (2.9)$$

$$\frac{\partial \psi_{j+1}(x_0^{(j+1)}, t_0, \omega)}{\partial x} = \theta_{j+1}, \quad 0 \leq j \leq r-1, \quad (2.10)$$

where $x_0^{(j+1)}$ is the point of reflection of γ_j at $\partial\Omega'(t_0)$ and θ_{j+1} is the direction of γ_{j+1} . We assume that γ_j , $1 \leq j \leq r$ do not contain caustic points. Then the eiconals ψ_j , $1 \leq j \leq r$ exist. In particular, this is true when $\Omega_j(t)$ are convex. Functions a_{pj} satisfy the following equations:

$$2 \frac{\partial a_{pj}}{\partial x} \cdot \frac{\partial \psi_j}{\partial x} + \Delta \psi_j a_{pj} - 2iA(x, t) \cdot \frac{\partial \psi_j}{\partial x} a_{pj} = f_{pj}(x, t, \omega) + i \frac{\partial \psi_j}{\partial t} a_{pj}, \quad p \geq 0, \quad (2.11)$$

where $f_{0j}=0$, f_{pj} depends on $a_{0j}, \dots, a_{p-1,j}$. When $\Omega'(t)$ is independent of t then ψ_j , $j \geq 1$ is also independent of t but $\psi_j(x, t, \omega)$ depends on t when $\Omega'(t)$ depends on t . We impose the following conditions on a_{pj} :

$$a_{pj}|_{\partial\Omega'(t)} = -a_{p,j+1}|_{\partial\Omega'(t)}, \quad 0 \leq j \leq r-1. \quad (2.12)$$

This conditions imply that

$$u_N|_{\partial\Omega'(t)} = 0. \quad (2.13)$$

Substituting (2.7) into (1.1) we get

$$\left(i \frac{\partial}{\partial t} - H\right) u_N = f_N, \quad (2.14)$$

where $f_N = \sum_{j=0}^r (f_{Nj}/(ik)^N) e^{ik\psi_j - ik^2 t} f_{Nj}(x, t, \omega)$ are smooth.

In order to complete the construction of the geometric optics solutions we need the following lemma.

Lemma 2.1: Consider the initial-boundary value problem

$$\begin{aligned} \left(i \frac{\partial}{\partial t} - H\right) u &= f \quad \text{in } D, \\ u|_{\partial D_t} &= 0, \quad \forall t \in (0, T), \end{aligned} \quad (2.15)$$

$$u(x, 0) = u_0(x), \quad (x, 0) \in D_0.$$

Suppose that $\Omega'(t)$ is a smooth domain in $\mathbf{R}^n$ for $\forall t \in [0, T]$. Then for any $f \in C^1([0, T], L^2(D_t))$ and any $u_0 \in H^2(D_0) \cap \dot{H}^1(D_0)$ there exists a unique solution $u \in C([0, T], \dot{H}^1(D_t) \cap H^2(D_t)) \cap C^1([0, T], L^2(D_t))$.

Here $H^p(D_t)$ is the Sobolev space with the norm $\|u\|_p^2 = \sum_{|k|=0}^p \int_{D_t} |\partial^k u|^2 dx$, $p=1, 2$, $\dot{H}^1(D_t)$ is the subspace of $H^1(D_t)$ consisting of functions equal to zero on ∂D_t .

Proof: We assume that $A(x, t)$ are real but $V(x, t)$ can be complex valued.

Make change of the variables:

$$t = t, \quad x' = \varphi(x, t), \quad \varphi(x, t) = (\varphi_1(x, t), \dots, \varphi_n(x, t)) \quad (2.16)$$

in $\bar{D}$ such that $\varphi(x, t) = x$ on $\partial\Omega_0 \times [0, T]$ and $\varphi(\Omega_j(t), t) = \Omega_j(0)$, $\forall t \in [0, T]$. Denote $\hat{u}(x', t) = u(x, t)$, where $x' = \varphi(x, t)$. Then $u_t(x, t) = \hat{u}_t + \hat{u}_{x'}(x', t) \cdot \varphi_t(x, t)$, $u_x(x, t) = u_{x'}(\partial\varphi/\partial x)$, etc.

Therefore (2.15) has the following form in (x', t) coordinates:

$$\begin{aligned} i \frac{\partial \hat{u}}{\partial t} &= \hat{H}_0 \hat{u} + \hat{H}_1 \hat{u} + \hat{f}, \quad (x', t) \in \hat{D}, \\ \hat{u}|_{\partial\hat{\Omega} \times (0, T)} &= 0, \quad \hat{u}(x', 0) = \hat{u}_0(x'), \quad x' \in \hat{\Omega}, \end{aligned} \quad (2.17)$$

where

$$\hat{H}_0 = \frac{1}{\sqrt{g(x', t)}} \sum_{j,k=1}^n \left(-i \frac{\partial}{\partial x'_j} - \hat{A}_j(x', t) \right) \sqrt{g} g^{jk}(x', t) \left(-i \frac{\partial}{\partial x'_k} - \hat{A}_k \right),$$

$$H_1 \hat{u} = \hat{V}(x', t) \hat{u} - i \varphi_t \cdot \hat{u}_{x'},$$

$$g^{jk}(x', t) = \sum_{p=1}^n \frac{\partial \varphi_j}{\partial x_p} \frac{\partial \varphi_k}{\partial x_p}, \quad g(x', t) = \det \|g^{jk}\|^{-1},$$

$$A_k(x, t) = \sum_{j=1}^n \hat{A}_j(x', t) \frac{\partial \varphi_j(x)}{\partial x_k}, \quad 1 \leq k \leq n.$$

Note that $\hat{D} = \hat{\Omega} \times (0, T]$, $\hat{\Omega} = \Omega_0 \setminus \bar{\Omega}'$, where $\Omega' = \cup_{j=1}^m \Omega_j(0)$.

Multiplying (2.17) by $\sqrt{g(x', t)} \bar{\hat{u}}(x', t)$ and integrating over $\hat{\Omega}$ we get

$$\left(i \frac{\partial \hat{u}}{\partial t}, \hat{u} \right)_g = (\hat{H}_0 \hat{u}, \hat{u})_g + (\hat{H}_1 u + f, \hat{u})_g, \quad (2.18)$$

where $(v_1, v_2)_g$ is the L_2 inner product in $\hat{\Omega}$ with the weight $\sqrt{g}$. Take the imaginary part of (2.18) and integrate in t from 0 to t . Since $\hat{H}_0$ is self-adjoint and since

$$\int_{\hat{\Omega}} i \sum_j \varphi_{jt} \frac{\partial \hat{u}}{\partial x_j} \sqrt{g} \bar{\hat{u}} dx' = - \int_{\hat{\Omega}} i \sum_j \varphi_{jt} \sqrt{g} \hat{u} \frac{\partial \hat{u}}{\partial x_j} dx' + \int_{\hat{\Omega}} b(x', t) |\hat{u}|^2 dx' \quad (2.19)$$

for some $b(x', t)$, we get

$$\int_{\hat{\Omega}} \sqrt{g(x', t)} |\hat{u}(x', t)|^2 dx' - \int_{\hat{\Omega}} \sqrt{g(x', 0)} |\hat{u}_0(x')|^2 dx' \leq C \int_0^t \|\hat{u}\|_0^2 dt' + C \int_0^t \|\hat{f}\|_0 \|\hat{u}\|_0 dt', \quad (2.20)$$

where $\|\hat{u}\|_0$ is the L_2 norm in $\hat{\Omega}$.

It follows from (2.20) by Gronwall's inequality that

$$\max_{[0, T]} \|\hat{u}(\cdot, t)\|_0^2 \leq C \|\hat{u}(x', 0)\|_0^2 + C \int_0^T \|\hat{f}(\cdot, t)\|_0^2 dt'. \quad (2.21)$$

Changing from the beginning in (1.1) u to $e^{i\lambda t} u$ we can assume that $\Re V(x, t)$ is large. Since $\hat{u}|_{\partial \hat{\Omega} \times (0, T)} = 0$ and $\hat{H}_0 + \hat{H}_1$ is elliptic for each $t \in [0, T]$ we get from (2.17), by the standard elliptic theory,

$$\|\hat{u}\|_2 \leq C \left\| \frac{\partial \hat{u}}{\partial t} \right\| + C \|\hat{f}\|_0, \quad (2.22)$$

where $\|\hat{u}\|_2$ is the norm in $H^2(\hat{\Omega})$.

Differentiate (2.17) in t ,

$$i \frac{\partial^2 \hat{u}}{\partial t^2} = \hat{H}_0 \frac{\partial \hat{u}}{\partial t} + \hat{H}_1 \frac{\partial \hat{u}}{\partial t} + \hat{H} \hat{u} + \frac{\partial \hat{f}}{\partial t}, \quad (2.23)$$

where $\hat{H}'$ is a differential operator in x of order at most 2. Note that $\partial \hat{u} / \partial t|_{\partial \hat{\Omega} \times [0, T]} = 0$ since $\hat{u}|_{\partial \hat{\Omega} \times [0, T]} = 0$. Multiplying (2.23) by $\sqrt{g}(\partial \hat{u} / \partial t)$, integrating over $\hat{\Omega} \times (0, t)$, and taking the imaginary part we get, as in (2.20),

$$\begin{aligned} \int_{\hat{\Omega}} \sqrt{g} \left| \frac{\partial \hat{u}}{\partial t} \right|^2 dx' - \int_{\hat{\Omega}} \sqrt{g(x', 0)} \left| \frac{\partial \hat{u}(x', 0)}{\partial t} \right|^2 dx' \leq C \int_0^t \left\| \frac{\partial \hat{u}}{\partial t} \right\|_0^2 dt' + C \int_0^t \|\hat{u}\|_2 \left\| \frac{\partial \hat{u}}{\partial t} \right\|_0 dt' \\ + C \int_0^t \left\| \frac{\partial \hat{f}}{\partial t} \right\|_0 \left\| \frac{\partial \hat{u}}{\partial t} \right\|_0 dt'. \end{aligned} \quad (2.24)$$

Using (2.22) we get from (2.24)

$$\max_{[0, T]} \left\| \frac{\partial \hat{u}}{\partial t} \right\|_0^2 \leq C \left\| \frac{\partial \hat{u}(\cdot, 0)}{\partial t} \right\|_0^2 + C \int_0^T \left(\|\hat{f}\|_0^2 + \left\| \frac{\partial \hat{f}}{\partial t} \right\|_0^2 \right) dt'. \quad (2.25)$$

It follows from (2.17) that

$$\left\| \frac{\partial \hat{u}(x', 0)}{\partial t} \right\|_0 \leq C \|\hat{u}_0(x', 0)\|_2 + \|\hat{f}(x', 0)\|_0. \quad (2.26)$$

Combining (2.21), (2.22), (2.25), and (2.26) we get

$$\max_{[0, T]} \|\hat{u}\|_2^2 + \max_{[0, T]} \left\| \frac{\partial \hat{u}}{\partial t} \right\|_0^2 \leq C \|\hat{u}_0\|_2^2 + C \max_{[0, T]} \|\hat{f}\|_0^2 + C \int_0^T \left\| \frac{\partial \hat{f}}{\partial t} \right\|_0^2 dt'. \quad (2.27)$$

We proved estimate (2.27) assuming that the solution $\hat{u}$ exists. To prove the existence we shall use the parabolic regularization.

Consider the parabolic equation in $\hat{D}$ as follows:

$$(i - \varepsilon) \frac{\partial u_\varepsilon}{\partial t} = \hat{H}_0 u_\varepsilon + \hat{H}_1 u_\varepsilon + \hat{f}, \quad \varepsilon > 0, \quad (2.28)$$

with the same boundary and initial conditions as in (2.17),

$$u_\varepsilon|_{\partial \hat{\Omega} \times (0, T)} = 0, \quad u_\varepsilon(x', 0) = \hat{u}_0(x'). \quad (2.29)$$

By the parabolic theory there exists a solution of (2.28) and (2.29) belonging to the same space as $\hat{u}$. Note that the estimate (2.27) holds for u_ε with constants independent of $\varepsilon > 0$. Taking the weak limit of a subsequence $\varepsilon_k \rightarrow 0$ we get the existence of $\hat{u}$ satisfying (2.17). Note that in the original coordinates $x = \varphi^{-1}(x', t)$ the estimate (2.27) still holds. ■

Remark 2.1: In the case of several obstacles in $\mathbf{R}^2$ we assume that $\Omega_j(t)$, $1 \leq j \leq m$ are convex sets having corners. In this case the estimate (2.22) does not hold and one has a weaker estimate as follows:

$$\|\hat{u}\|_1 \leq C \left\| \frac{\partial \hat{u}}{\partial t} \right\|_0 + C \|\hat{f}\|_0. \quad (2.30)$$

To prove the existence of the solution of (2.17) in this case we assume, for simplicity, that obstacles $\Omega_j(t)$ move as rigid bodies, i.e., the change of variables (2.16) has the form $t = t$, $\varphi(x, t) = x + \varphi_j(t)$, $\varphi_j(0) = 0$, near $\Omega_j(t)$, $1 \leq j \leq m$. Then $g^{jk}(x', t) = \delta_{jk}$ near Ω' and the differential operator $\hat{H}'$ has the order 1 near Ω' . Therefore (2.24) holds with $\|\hat{u}\|_2$ replaced by $\|\hat{u}\|_1 + \|\chi \hat{u}\|_2$ where $\chi = 0$ near Ω' . Using the elliptic estimate of the form (2.22) for $\chi \hat{u}$ and the estimate (2.30) we get that (2.25) holds in the case of corners too. Using the parabolic regulation as in (2.28) we get that there exists a unique solution of (2.15) satisfying the estimate (2.27) with $\|\hat{u}\|_2$ replaced by $\|\hat{u}\|_1$. ■

By Lemma 2.1 and Remark 2.1 there exists $u^{(N+1)} \in C([0, T], H^1(D_t)) \cap C^1([0, T], L^2(D_t))$ such that

$$\left(i\frac{\partial}{\partial t} - H\right)u^{(N+1)} = -f_N \quad \text{in } D,$$

$$u^{(N+1)}|_{\partial\Omega'(t)} = 0, \quad \forall t \in [0, T],$$

$$u^{(N+1)}(x, 0) = 0, \quad u^{(N+1)}|_{\partial\Omega_0 \times (0, T)} = 0,$$
(2.31)

where f_N is the same as in (2.14). Then $\|u^{(N+1)}\|_1 + \|\partial u^{(N+1)} / \partial t\|_0 \leq Ck^{-N+2}$ for $\forall t \in [0, T]$. Therefore $u = u_N + u^{(N+1)}$ solves

$$\left(i\frac{\partial}{\partial t} - H\right)u = 0, \quad (x, t) \in D, \quad u|_{\partial\Omega_0 \times (0, T)} = u_N|_{\partial\Omega_0 \times (0, T)},$$

$$u|_{\partial\Omega'(t)} = 0, \quad \forall t \in [0, T], \quad u(x', 0) = 0, \quad (x', 0) \in D_0.$$
(2.32)

Now we shall use Green's formula. Suppose we are given two time-dependent Schrödinger operators $i(\partial/\partial t) - H_1$ and $i(\partial/\partial t) - H_2$ such that the corresponding D-to-N operators Λ_1 and Λ_2 are gauge equivalent, i.e., $\Lambda_1 = g_0^{-1} \Lambda_2 g_0$ where $g_0 = g|_{\partial\Omega_0 \times [0, T]}, g \in G(D)$. Denote $A^{(3)} = A^{(2)} - ig^{-1}(\partial g / \partial x)$, $V^{(3)} = V^{(2)} + ig^{-1}(\partial g / \partial t)$. Let $i(\partial/\partial t) - H_3$ be the Schrödinger operator with electromagnetic potentials $A^{(3)}$ and $V^{(3)}$. Then $i(\partial/\partial t) - H_1$ and $i(\partial/\partial t) - H_3$ have the same D-to-N operator: $\Lambda_3 = \Lambda_1$.

Let u_1 be the solution of the initial-boundary value problem

$$i\frac{\partial u_1}{\partial t} - H_1 u_1 = 0 \quad \text{in } D,$$

$$u_1(x, 0) = 0, \quad (x, 0) \in D_0,$$
(2.33)

$$u_1|_{\partial\Omega_0 \times (0, T)} = f, \quad u_1|_{\partial\Omega'(t)} = 0 \quad \forall t \in [0, T],$$

and let u_3 be the solution of

$$i\frac{\partial u_3}{\partial t} - H_3^* u_3 = 0 \quad \text{in } D,$$

$$u_3(x, T) = 0, \quad (x, T) \in D_T,$$
(2.34)

$$u_3|_{\partial\Omega_0 \times (0, T)} = g, \quad u_3|_{\partial\Omega'(t)} = 0 \quad \forall t \in [0, T],$$

where H_3^* is the adjoint to H_3 .

We have, by Green's formula (cf. Ref. 18),

$$0 = \left(\left(i\frac{\partial u}{\partial t} - H_1 \right) u_1, u_3 \right) - \left(u_1, \left(i\frac{\partial}{\partial t} - H_3^* \right) u_3 \right) = - \int_D \left(2iA^{(1)}(x, t) \cdot \frac{\partial u_1}{\partial x} \overline{u_3} + 2iu_1 A^{(3)} \cdot \frac{\partial \overline{u_3}}{\partial x} + (q_1(x, t) \right.$$

$$\left. - q_3(x, t)) u_1 \overline{u_3} \right) dx dt + [\Lambda_1 f, g] - [f, \Lambda_3^* g],$$
(2.35)

where $q_p = i(\partial/\partial x) \cdot A^{(p)}(x, t) + A^{(p)}(x, t) \cdot A^{(p)}(x, t) + V^{(p)}(x, t)$, $[\cdot, \cdot]$ is the inner product on $\partial\Omega_0 \times [0, T]$, Λ_p are the D-to-N operators corresponding to H_p , $p=1$ or 3 , and Λ_3^* is the adjoint to Λ_3 . Let

$$v = \sum_{j=0}^r \sum_{p=0}^N \frac{b_{pj}}{(ik)^p} e^{-ik^2 t + ik \psi_j(x,t,\omega)} + v^{(N+1)} \quad (2.36)$$

be the geometric optics solution of (2.34) corresponding to the same broken ray γ as the solution (2.32). Substitute (2.32) in (2.35) instead of u_1 and substitute (2.36) instead of u_3 . Dividing (2.35) by $2k$, passing to the limit when $k \rightarrow \infty$, and taking into account that $\Lambda_1 = \Lambda_3$ we get (cf. Ref. 6 pp. 1502–1503)

$$0 = \sum_{j=0}^r \int_D (A^{(3)}(x,t) - A^{(1)}(x,t)) \cdot \psi_{jx}(x,t,\theta) a_{0j}(x,t,\omega) \overline{b_{0j}(x,t,\omega)} dx dt.$$

Note that a_{00} and b_{00} have the form (2.4) and a_{0j} and b_{0j} have the form (2.21) in Ref. 6, p. 1502, with factor $\exp(i \int_0^t \psi_j d\tau)$ added. Note also that terms containing ψ_{jt} in $a_{0j} \overline{b_{0j}}$ cancel each other. Making changes of variables as in Ref. 6, pp. 1503–1506 (see also Ref. 7, p. 30), and taking the limit when $\varepsilon \rightarrow 0$ where $\varepsilon > 0$ is the same as in (2.4), we obtain

$$\exp \left[i \sum_{j=0}^r \int_{\gamma_j} (A^{(3)}(x_0^{(j)} + s \theta_j, t_0) - A^{(1)}(x_0^{(j)} + s \theta_j, t_0)) \cdot \theta_j ds \right] = 1, \quad (2.37)$$

where θ_j is the direction of γ_j , $x_0^{(j)}$ is the starting point of γ_j , $0 \leq j \leq r$.

It is important to emphasize that the ray γ is contained in the plane $t = \text{const}$. Therefore the tomography problem (2.37) is exactly the same as in the time-independent case.

In the case when γ does not hit obstacles, (2.37) reduces to

$$\exp \left\{ i \int_{-\infty}^{\infty} [A^{(3)}(x_0^{(0)} + s \omega, t_0) - A^{(1)}(x_0^{(0)} + s \omega, t_0)] \cdot \omega ds \right\} = 1. \quad (2.38)$$

We took in (2.38) $s_0 = -\infty$, $s_1 = +\infty$ since $A^{(j)}$, $j=1,3$ have compact support.

If conditions of Theorem 1.1 are satisfied then (2.38) implies (see Ref. 5, Sec. II) that there exists $\varphi(x,t) \in C^\infty(\bar{D})$ such that

$$A^{(1)}(x,t) - A^{(3)}(x,t) = \frac{\partial \varphi}{\partial x}, \quad \varphi|_{\partial \Omega_0 \times [0,T]} = 0. \quad (2.39)$$

If conditions of Theorem 1.2 are satisfied then (2.37) implies (see Ref. 6, pp. 1512–1515) that there exists $c(x,t) \in G(D)$ such that

$$A^{(3)} - A^{(1)} = ic^{-1} \frac{\partial c}{\partial x}, \quad c|_{\partial \Omega_0 \times [0,T]} = 1. \quad (2.40)$$

Using (2.39) or (2.40) we make a gauge transformation $u_4 = c^{-1} u_3$, where $c = e^{i\varphi}$ in the case (2.39). Then $(i(\partial/\partial t) - H_4)u_4 = 0$, where $i(\partial/\partial t) - H_4$ is the operator with electromagnetic potentials $A^{(4)} = A^{(3)} - ic^{-1}(\partial c/\partial x)$, $V^{(4)} = V^{(3)} + ic^{-1}(\partial c/\partial t)$. Note that $A^{(1)} = A^{(4)}$. Since $c=1$ on $\partial \Omega_0 \times (0,T)$ we have that $\Lambda_4 = \Lambda_3 = \Lambda_1$, where Λ_4 is the D-to-N operator corresponding to $i(\partial/\partial t) - H_4$. Apply Green's formula (2.35) to $(i(\partial/\partial t) - H_1)u_1 = 0$ and $(i(\partial/\partial t) - H_4^*)u_5 = 0$, where $u_5(x,T) = 0$, $u_5|_{\partial \Omega_0 \times (0,T)} = g_1$, $u_5|_{\partial \Omega'(t)} = 0$, $\forall t \in [0,T]$.

Since $A^{(1)} = A^{(4)}$ and $\Lambda_1 = \Lambda_4$ we get

$$\int_D \left(V^{(1)}(x,t) - V^{(3)}(x,t) - ic^{-1} \frac{\partial c}{\partial t} \right) u_1 \overline{u_5} dx dt = 0. \quad (2.41)$$

Substituting geometric optic solutions (2.32) and (2.36) with H_3^* replaced by H_4^* we get

$$\sum_{j=1}^r \int_{\gamma_j} \left(V^{(1)} - V^{(3)} - ic^{-1} \frac{\partial c}{\partial t} \right) ds = 0. \quad (2.42)$$

Note that (2.42) holds for each $t \in [0, T]$ and for each broken ray in D_t . It is shown in Ref. 6, Sec. III that (2.42) implies that

$$V^{(1)} = V^{(3)} + ic^{-1} \frac{\partial c}{\partial t}. \quad (2.43)$$

Therefore (2.40) and (2.43) prove Theorem 1.2.

Analogously, when γ does not hit the obstacles we have, instead of (2.42),

$$\int_{\gamma} \left(V^{(1)} - V^{(3)} - ic^{-1} \frac{\partial c}{\partial t} \right) ds = 0. \quad (2.44)$$

It is shown in Ref. 5, Sec. II that (2.44) also implies (2.43) with $c = e^{i\varphi(x,t)}$. Therefore Theorem 1.1 also holds.

Remark 2.2: Deriving formulas of the form (2.37) we assumed that $\gamma = \gamma_1 \cup \dots \cup \gamma_r$ are broken nontangential rays that do not have focal points (i.e., do not intersect the caustics set). It was noted in Ref. 7, p. 29 that these formulas still hold when γ contains a generic caustics point.

Consider, for simplicity, the case $n=2$. Let (x_{10}, x_{20}, t_0) be the point of the intersection of γ with the caustics set. In the neighborhood of (x_{10}, x_{20}, t_0) we replace the ansatz $\sum_{p=0}^N (a_p(x, t) / (ik)^p) e^{-ik^2 t + ik\varphi(x, t)}$ with the following ansatz (cf. Ref. 19):

$$v = \sum_{p=0}^N \int_{-\infty}^{\infty} \frac{b_p(x, t, \xi_1)}{(ik)^{p-1/2}} e^{-ik^2 t + i\varphi(x_2, t, \xi_1) - ix_1 \xi_1} d\xi_1. \quad (2.45)$$

At the caustics set we have

$$\varphi_{\xi_1} - x_1 = 0, \quad \varphi_{\xi_1}^2 = 0.$$

We shall assume that $\varphi_{\xi_1^3} \neq 0$ on this set. Then the following estimate holds in a neighborhood of (x_{10}, x_{20}, t_0) .

$$|v| \leq \frac{Ck^{1/6}}{1 + k^{1/6}d^{1/4}} \leq \frac{C}{d^{1/4}}, \quad (2.46)$$

where $d(x, t)$ is the distance to the caustics set. It was shown in Ref. 7 that the estimate (2.46) leads to the proof of (2.37) and (2.42).

Remark 2.3: Consider the case when $\Omega_0 \subset \mathbb{R}^n$, $n \geq 2$ is a domain containing finite number of smooth obstacles Ω_j , $1 \leq j \leq m$, $V=0$ and the magnetic field B is time independent. Suppose $B = \text{curl } A$ is shielded inside the obstacles Ω_j , $1 \leq j \leq m$, i.e., $B=0$ in $D = \Omega_0 \setminus \bigcup_{j=1}^m \Omega_j$. Then $\int_{\gamma} A \cdot dx$ depends only on the homotopy class of path γ in D . If we can find a finite number of broken rays γ_i , $i=1, \dots, N$ that generate the homotopy group of D , then $\alpha_i = \int_{\gamma_i} A \cdot dx$, $i=1, \dots, N$ determine the gauge equivalence class of A , i.e., we can solve the inverse problem using only finite number of boundary measurements.

III. THE SCHRÖDINGER OPERATOR WITH TIME-DEPENDENT YANG–MILLS POTENTIALS

The Schrödinger equation with time-dependent Yang–Mills potentials has the following form:

$$Lu \stackrel{\text{def}}{=} i \frac{\partial u(x,t)}{\partial t} - \sum_{j=1}^n \left(-i I_m \frac{\partial}{\partial x_j} - A_j(x,t) \right)^2 u(x,t) + V(x,t)u(x,t), \quad (3.1)$$

where A_j , $1 \leq j \leq n$, $V(x,t)$, $u(x,t)$ are $m \times m$ matrices, and I_m is the identity matrix in $\mathbf{C}^m$. We assume that A_j , $1 \leq j \leq n$, V are smooth in $\bar{\Omega}_0 \times [0, T]$ and without loss of generality (see Remark 3.1) we can assume that $A(x,t)=0$, $V(x,t)=0$ near $\partial\Omega_0 \times (0, T)$. We also assume that $A_j(x,t)$ are self-adjoint matrices, $1 \leq j \leq n$. We consider (3.1) in $\Omega_0 \times (0, T)$ with the initial condition (1.2) and the boundary condition $u|_{\partial\Omega_0 \times (0, T)} = f$. We do not consider in this paper domains with obstacles in the case of Eq. (3.1).

The gauge group G now is the group of $m \times m$ nonsingular smooth matrices $g(x,t)$ in $\bar{\Omega}_0 \times [0, T]$. Let $L^{(p)}$ be two Schrödinger operators of the form (3.1) with Yang–Mills potentials $A^{(p)}(x,t)$, $V^{(p)}(x,t)$, $p=1, 2$.

Let

$$\Lambda_p f_p = \left(I_m \frac{\partial}{\partial \nu} - i A \cdot \nu \right) u_p|_{\partial\Omega_0 \times (0, T)}, \quad p=1, 2 \quad (3.2)$$

be corresponding D-to-N operators, where $L^{(p)}u_p=0$ in $\Omega_0 \times (0, T)$, $u_p=0$ when $t=0$, $p=1, 2$, $u_p|_{\partial\Omega_0 \times (0, T)} = f_p$. If $u_2 = g^{-1}u_1$, $g \in G$, then

$$A_j^{(2)} = g^{-1}A_j^{(1)}g + ig^{-1}\frac{\partial g}{\partial x_j}, \quad 1 \leq j \leq n, \quad (3.3)$$

$$V^{(2)} = g^{-1}V^{(1)}g - ig^{-1}\frac{\partial g}{\partial t}.$$

Yang–Mills potentials $(A^{(2)}, V^{(2)})$ and $(A^{(1)}, V^{(1)})$ are called gauge equivalent. As in the case of Theorem 1.1 we have the following.

Theorem 3.1: *If the D-to-N operators Λ_1 and Λ_2 are gauge equivalent on $\partial\Omega_0 \times (0, T)$ [cf. (1.7)] then $(A^{(1)}, V^{(1)})$ and $(A^{(2)}, V^{(2)})$ are gauge equivalent in $\bar{\Omega}_0 \times [0, T]$.*

The start of the proof of Theorem 3.1 is the same as of Theorem 1.1. We construct the geometric optics solution of (3.1) of the form

$$u_N = e^{-ik^2 t + ik(x, \omega)} \sum_{p=0}^N \frac{1}{(ik)^p} a_p(x, t, \omega) + u^{(N+1)}, \quad (3.4)$$

where $a_0(x, t) = \chi_1(t)\chi_2(\tau)c(x, t, \omega)$ and

$$\omega \cdot \frac{\partial c}{\partial x} - iA(x, t) \cdot \omega c(x, t) = 0, \quad s > s_0, \quad (3.5)$$

$$c|_{s=s_0} = I_m,$$

$a_p(x, t, \omega)$ are solutions of the equations $2\omega \cdot (-i(\partial/\partial x) - A)a_p = La_{p-1}$ for $s > s_0$, $a_p|_{s=s_0} = 0$, $p \geq 1$.

Here s , s_0 , $\chi_1(t)$, $\chi_2(\tau)$ are the same as in (2.4) and (2.5). Note that since $A(x, t)$ have a compact support we can take $s_0 = -\infty$. Finally, let $u^{(N+1)}$ satisfy

$$Lu^{(N+1)} = \frac{-e^{-ik^2 t + ik(\omega \cdot x)}}{(ik)^N} La_N, \quad (x, t) \in \Omega_0 \times (0, T),$$

$$u^{(N+1)}(x, 0) = 0, \quad u^{(N+1)}|_{\partial\Omega_0 \times (0, T)} = 0. \quad (3.6)$$

The existence of $u^{(N+1)}$ follows from Lemma 2.1 that holds without changes in the case of Eq. (3.1). Since $\Lambda_2 = g_0^{-1} \Lambda_1 g_0$ for some $g \in G(\Omega_0 \times (0, T))$ where $g_0 = g|_{\partial\Omega_0 \times (0, T)}$ we have that $\Lambda_3 = \Lambda_1$ where Λ_3 is the D-to-N operator corresponding to L_3 with potentials $A^{(3)} = g^{-1} A^{(2)} g - i g^{-1} (\partial g / \partial x)$, $V^{(3)} = g^{-1} V^{(2)} g + i g^{-1} (\partial g / \partial t)$. Substituting the geometric optics solutions for L_1 and L_3^* into Green's formula of the form (2.35) we get as in Sec. II [cf. (2.38) and also Ref. 7, pp. 29 and 30]

$$c_{30}^{-1}(+\infty, y', t, \omega) c_{10}(+\infty, y', t, \omega) = I_m, \quad (3.7)$$

where $y_1 = x \cdot \omega$, $y' = x - (x \cdot \omega) \omega$, $c_j(x, t, \omega)$, $j = 1, 3$ are the solution of the differential equation of the form [cf. (3.5)]

$$\omega \cdot \frac{\partial c_j(x, t, \omega)}{\partial x} - i(A_j(x, t) \cdot \omega) c_j(x, t, \omega) = 0, \quad (3.8)$$

$c_{j0}(y_1, y', t, \omega)$ is $c_j(x, t, \omega)$ in (y_1, y') coordinates, $c_{j0}(-\infty, y', t, \omega) = I_m$. The matrix $c_{j0}(+\infty, y', t, \omega)$ is called the non-Abelian Radon transform of $A^{(j)}(x, t)$, $j = 1, 3$ (cf. Refs. 15 and 8).

The following lemma was proven in Ref. 8 (see also Ref. 15).

Lemma 3.1: *If the non-Abelian Radon transforms of $A^{(1)}$ and $A^{(3)}$ are equal for all ω and t then $A^{(1)}$ and $A^{(3)}$ are gauge equivalent with $g \in G_0(\Omega_0 \times [0, T])$, i.e., $g = I_m$ on $\partial\Omega_0 \times [0, T]$.*

The proof of Lemma 3.1 is much harder than the similar result in the Abelian case (i.e., when $m = 1$), and involves a lot of the complex analysis. We consider briefly the most important case of two dimensions. The case $n \geq 3$ is reduced easily to the case $n = 2$. The main ingredient of the proof is the study of an equation of the form (3.8) with complex parameters

$$\zeta_1 \frac{\partial c}{\partial x_1} + \zeta_2 \frac{\partial c}{\partial x_2} = i(A_1(x, t) \zeta_1 + A_2(x, t) \zeta_2) c, \quad (3.9)$$

where $\zeta_1^2 + \zeta_2^2 = 1$, $\zeta_i \in \mathbf{C}$, $i = 1, 2$. Define $\zeta_1(\tau) = 1/2(\tau + 1/\tau)$, $\zeta_2(\tau) = i/2(\tau - 1/\tau)$, where $\tau \in \mathbf{C} \setminus \{0\}$. It was proven in Ref. 12 that there exists nonsingular matrix solutions $c_{\pm}(x, t, \tau)$ of (3.9), smooth in (x, t, τ) , where $|\tau| \leq 1$ in the case of c_+ and $|\tau| \geq 1$ in the case of c_- , and such that $c_+(x, t, \tau)$ is analytic in τ when $|\tau| < 1$, $c_-(x, t, \tau)$ is analytic in τ for $|\tau| > 1$. Let $\tau = |\tau| e^{i\varphi}$. When $|\tau| \rightarrow 1$ we have

that $(\zeta_1(\tau), \zeta_2(\tau)) \xrightarrow{\text{def}} \omega(\varphi) = (\cos \varphi, -\sin \varphi)$. Therefore $c_{\pm}(x, t, e^{i\varphi})$ are solutions of equation of the form (3.8) with $\omega(\varphi) = (\cos \varphi, -\sin \varphi)$. Denote $y_1 = x \cdot \omega(\varphi)$ and $y_2 = x \cdot \omega_{\perp}(\varphi)$ where $\omega_{\perp}(\varphi) = (\sin \varphi, \cos \varphi)$. Let $c_{\pm}^{(0)}(y_1, y_2, t, e^{i\varphi})$ be $c_{\pm}(x, t, e^{i\varphi})$ in (y_1, y_2) coordinates and let $c_0(y_1, y_2, t, \omega(\varphi))$ be the solution of the Cauchy problem (3.8). We have

$$c_0(y_1, y_2, t, \omega(\varphi)) = c_{\pm}^{(0)}(y_1, y_2, t, e^{i\varphi}) (c_{\pm}^{(0)}(-\infty, y_2, t, e^{i\varphi}))^{-1} \quad (3.10)$$

by the uniqueness of the Cauchy problem for (3.8). Putting $y_1 = +\infty$ we get

$$c_0(+\infty, y_2, t, \omega(\varphi)) = c_{\pm}^{(0)}(+\infty, y_2, t, e^{i\varphi}) (c_{\pm}^{(0)}(-\infty, y_2, t, e^{i\varphi}))^{-1}. \quad (3.11)$$

It follows from (3.7) and (3.11) that

$$c_{1,\pm}^{(0)}(+\infty, y_2, t, e^{i\varphi}) (c_{1,\pm}^{(0)}(-\infty, y_2, t, e^{i\varphi}))^{-1} = c_{3,\pm}^{(0)}(+\infty, y_2, t, e^{i\varphi}) (c_{3,\pm}^{(0)}(-\infty, y_2, t, e^{i\varphi}))^{-1}, \quad (3.12)$$

where $c_{j,\pm}(x, t, e^{i\varphi})$ correspond to the potentials $A^{(j)}(x, t)$, $j = 1, 3$. The continuation of the proof of Lemma 3.1 when $n = 2$ is the same as in Ref. 8, Sec. II.

Now we shall consider the case $n \geq 3$ assuming that (3.7) holds. Let $\omega = (\omega_1, \dots, \omega_n) \in S^{n-1}$ and let $\omega_{12} = (\omega_1, \omega_2, 0, \dots, 0) \in S^{n-1}$. Consider Eq. (3.8) with ω replaced by ω_{12} . It is a two-dimensional problem with $x_3, \dots, x_n$ and t as parameters. It follows from Lemma 3.1 for $n = 2$ that there exists $g(x, t)$, $\det g(x, t) \neq 0$, $g = I_m$ on $\partial\Omega_0 \times [0, T]$ such that

$$A_1^{(3)} = g^{-1}A_1^{(1)}g + ig^{-1}\frac{\partial g}{\partial x_1}, \quad A_2^{(3)} = g^{-1}A_2^{(1)}g + ig^{-1}\frac{\partial g}{\partial x_2} \quad (3.13)$$

for all (x, t) . Replacing ω_2 by ω_j , $j > 2$, we get that there exists $\tilde{g} \in G_0(\Omega_0 \times [0, T])$ such that

$$A_1^{(3)} = \tilde{g}^{-1}A_1^{(1)}\tilde{g} + i\tilde{g}^{-1}\frac{\partial \tilde{g}}{\partial x_1}, \quad A_j^{(3)} = \tilde{g}^{-1}A_j^{(1)}\tilde{g} + i\tilde{g}^{-1}\frac{\partial \tilde{g}}{\partial x_j} \quad (3.14)$$

for all (x, t) . Comparing (3.13) and (3.14) we have that g and $\tilde{g}$ satisfy the same system of differential equations

$$i\frac{\partial g}{\partial x_1} + A_1^{(1)}g - gA_1^{(3)} = 0, \quad g|_{\partial\Omega_0 \times [0, T]} = I_m$$

for each $(x_2, \dots, x_n, t)$.

By the uniqueness theorem for the differential equations we get that $\tilde{g} = g$. Therefore

$$A_j^{(3)} = g^{-1}A_j^{(1)}g + ig^{-1}\frac{\partial g}{\partial x_j}, \quad j \geq 3. \quad (3.15)$$

Denote

$$A^{(4)} = gA^{(3)}g^{-1} - i\frac{\partial g}{\partial x}g^{-1}, \quad V^{(4)} = gV^{(3)}g^{-1} + i\frac{\partial g}{\partial x}g^{-1}. \quad (3.16)$$

Then $A^{(4)}(x, t) = A^{(1)}(x, t)$.

Now we shall show that $V^{(4)} = V^{(1)}$, i.e., $(A^{(3)}, V^{(3)})$ and $(A^{(1)}, V^{(1)})$ are gauge equivalent. As before, it is enough to consider the case $n=2$. Using Green's formula for $L^{(1)}$ and $L^{(4)*}$ we get, as in Sec. II [cf. (2.41) and (2.44)]

$$\int_{-\infty}^{\infty} c_{10}^{-1}(y_1, y_2, t, \omega)(V^{(1)}(x, t) - V^{(4)}(x, t))c_{10}(y_1, y_2, t, \omega)dy_1 = 0. \quad (3.17)$$

Here $y_1 = x \cdot \omega$, $y_2 = x \cdot \omega_{\perp}$. Substituting (3.10) into (3.17), and multiplying (3.17) by $c_{\pm}^{-1}(-\infty, y_2, t, e^{i\varphi})$ from the left and $c_{\pm}(-\infty, y_2, t, e^{i\varphi})$ from the right, we get

$$\int_{-\infty}^{\infty} c_{\pm}^{-1}(y_1, y_2, t, e^{i\varphi})(V^{(1)} - V^{(4)})c_{\pm}(y_1, y_2, t, e^{i\varphi})dy_1 = 0. \quad (3.18)$$

The continuation of proof that (3.18) leads to $V^{(1)} - V^{(4)} = 0$ is the same as in Ref. 4, pp. 61 and 62. ■

Remark 3.1: In this remark we shall show that without loss of generality one can assume that $A(x, t)$ and $V(x, t)$ are smooth and equal to zero near $\partial\Omega_0 \times [0, T]$ (cf. Refs. 16, 11, and 4). Let L_p be two operators in $\Omega_0 \times [0, T]$ such that $\Lambda_1 = \Lambda_2$ on $\partial\Omega_0 \times (0, T)$. As in Ref. 4, pp. 53 and 54, one can replace $(A^{(2)}, V^{(2)})$ in $\Omega_0 \times [0, T]$ by gauge equivalent $(A^{(3)}, V^{(3)})$ with the gauge $g \in G_0$ such that $A^{(3)}(x, t) \cdot n(x, t) = A^{(1)}(x, t) \cdot n(x, t)$ near $\partial\Omega_0 \times [0, T]$, where $n(x, t)$ is a smooth extension near $\partial\Omega_0 \times [0, T]$ of the unit normal vector $n(x)$ on $\partial\Omega_0$. We shall show that $\Lambda_1 = \Lambda_3$ on $\partial\Omega_0 \times (0, T)$ implies that $A^{(1)} = A^{(3)}$ and $V^{(1)} = V^{(3)}$ on $\partial\Omega_0 \times [0, T]$ and the same is true for its derivatives: Introduce in a neighborhood U of an arbitrary point $(x_0, t_0) \in \partial\Omega_0 \times [0, T_0]$ a system of coordinates (y, t) such that the equation of $\partial\Omega_0$ is $y_n = 0$. Let $(\eta_1, \dots, \eta_n, \eta_0)$ be the dual coordinates to (y, t) in the cotangent space $T^*(U \times [0, T])$. The Schrödinger operators L_1 and L_3 are elliptic microlocally in the region where $|\eta_0| < \varepsilon \sum_{k=1}^n \eta_k^2$ (cf. Ref. 9, Remark 2.2). Therefore the elliptic parametrix for Λ_1 and Λ_3 applies in this region. As in Ref. 4, Proposition 2.1, we get that

$$\frac{\partial^{p+q} A^{(1)}}{\partial x^p \partial t^q} = \frac{\partial^{p+q} A^{(3)}}{\partial x^p \partial t^q} \text{ and } \frac{\partial^{p+q} V^{(1)}}{\partial x^p \partial t^q} = \frac{\partial^{p+q} V^{(3)}}{\partial x^p \partial t^q}$$

on $\partial\Omega_0 \times (0, T)$ for any p and q .

Therefore we can extend smoothly $(A^{(1)}, V^{(1)})$ and $(A^{(3)}, V^{(3)})$ from $\bar{\Omega}_0 \times [0, T]$ to a large domain $\tilde{\Omega}_0 \times [0, T]$ in such a way that $A^{(1)} = A^{(3)}$, $V^{(1)} = V^{(3)}$ in $(\tilde{\Omega}_0 \setminus \Omega_0) \times [0, T]$ and $(A^{(1)}, V^{(1)}) = (A^{(3)}, V^{(3)}) = (0, 0)$ near $\partial\tilde{\Omega}_0 \times [0, T]$. It remains to show that $\Lambda_1 = \Lambda_3$ on $\partial\Omega_0 \times (0, T)$ implies that $\Lambda_1 = \Lambda_3$ on $\partial\tilde{\Omega}_0 \times [0, T]$. Take any smooth f on $\partial\tilde{\Omega}_0 \times [0, T]$ and consider the initial-boundary value problem

$$L^{(1)} u_1 = 0 \quad \text{in } \tilde{\Omega}_0 \times (0, T),$$

$$u_1(x, 0) = 0, \quad u_1|_{\partial\tilde{\Omega}_0 \times [0, T]} = f.$$

Let f_1 be the restriction of u_1 to $\partial\Omega_0 \times (0, T)$. Consider the initial-boundary value problem

$$L_3 u_3 = 0 \quad \text{in } \Omega_0 \times (0, T),$$

$$u_3(x, 0) = 0, \quad u_3|_{\partial\Omega_0 \times [0, T]} = f_1.$$

Let $w_3 = u_1$ in $(\tilde{\Omega}_0 \setminus \Omega_0) \times (0, T)$, $w_3 = u_3$ in $\Omega_0 \times (0, T)$. Since $\Lambda_1 = \Lambda_3$ on $\partial\Omega_0 \times (0, T)$ we get that w_3 is the solution of $L_3 w = 0$ in $\tilde{\Omega}_0 \times (0, T)$. Moreover, $w_3(x, 0) = 0$, $x \in \tilde{\Omega}_0$ and $w_3|_{\partial\tilde{\Omega}_0 \times [0, T]} = f$. Since $L^{(1)} = L^{(3)}$ and $u_1 = w_3$ in $(\tilde{\Omega}_0 \setminus \Omega_0) \times (0, T)$ we have that $\Lambda_1 = \Lambda_3$ on $\partial\tilde{\Omega}_0 \times (0, T)$. Therefore we have reduced the inverse problem in $\Omega_0 \times (0, T)$ to an equivalent inverse problem in a larger domain $\tilde{\Omega}_0 \times (0, T)$. Note that the same proof works when $\Omega_0 \times (0, T)$ contains obstacles.

IV. EXAMPLES OF THE GAUGE EQUIVALENCE CLASSES AND THE AHARONOV–BOHM EFFECT

The results of Sec. II are valid in the multiconnected domains D . In such domains there are electromagnetic potentials $(A(x, t), V(x, t))$ and $(A'(x, t), V'(x, t))$ that correspond to the same electromagnetic fields in D : $\text{curl } A = \text{curl } A' = B$, $E = -\partial V / \partial x - \partial A / \partial t = -\partial V' / \partial x - \partial A' / \partial t$; however, (A, V) and (A', V') are not gauge equivalent. There is a simple description of all gauge equivalent classes of potentials in D (cf. Refs. 23, 17, and 20) as follows.

Fix a point $(x_0, t_0) \in \partial\Omega_0 \times [0, T]$ and denote by $\mathcal{P}$ the group of all paths γ in $\bar{D}$ that start and end at (x_0, t_0) . Let $\int_\gamma A(x, t) \cdot dx - V(x, t) dt$ be a line integral in D . Denote

$$R(A, V, \gamma) = \exp \left\{ i \left(\int_\gamma -A(x, t) \cdot dx + V(x, t) dt \right) \right\}. \quad (4.1)$$

R is called the nonintegrable phase factor (cf. Refs. 23 and 17). $R(A, V, \gamma_0)$ is well defined for any closed path γ_0 in $\bar{D}$: Let γ_1 be a path connecting (x_0, t_0) with an arbitrary point $(x_1, t_1) \in \gamma_0$ and let $\gamma \in \mathcal{P}$ be a close path $\gamma_1 \cup \gamma_0 \cup \gamma_1^{-1}$. It is obvious that $R(A, V, \gamma) = R(A, V, \gamma_0)$. If (A, V) and (A', V') are gauge equivalent then

$$\begin{aligned} R' R^{-1} &= \exp \left\{ i \left(\int_\gamma (A - A') \cdot dx + (V' - V) dt \right) \right\} = \exp \left\{ -i \left(\int_\gamma ic^{-1} \frac{\partial c}{\partial x} \cdot dx + ic^{-1} \frac{\partial c}{\partial t} dt \right) \right\} \\ &= \exp \left\{ \int_\gamma d(\log c) \right\} = 1, \end{aligned}$$

since $c(x, t) \in G(D)$ is single valued in $\bar{D}$. Therefore $R' = R$; and vice versa, if $R(A, V, \gamma)$

$=R(A', V', \gamma)$ for all $\gamma \in \mathcal{P}$ then (A, V) and (A', V') are gauge equivalent: Let

$$\exp \left\{ i \left(\int_{\gamma} (A' - A) \cdot dx + (V - V') dt \right) \right\} = 1 \quad (4.2)$$

for all $\gamma \in \mathcal{P}$. Let (x, t) be an arbitrary point in $\bar{D}$. Denote by $c(x, t)$ the integral

$$c(x, t) = \exp \left\{ i \left(\int_{\gamma(x, t)} (A' - A) \cdot dx + (V - V') dt \right) \right\}, \quad (4.3)$$

where $\gamma(x, t)$ is an arbitrary path in $\bar{D}$ connecting (x_0, t_0) and (x, t) . It follows from (4.2) that (4.3) does not depend on $\gamma(x, t)$ connecting (x_0, t_0) and (x, t) . Differentiating (4.3) in x and t we get that (A, V) and (A', V') are gauge equivalent.

For fixed (A, V) $R(A, V, \gamma)$ defines a map of the group of paths $\mathcal{P}$ to the group U of complex numbers z with absolute value 1, i.e., $z = e^{i\alpha}$, $\alpha \in \mathbf{R}$ [we assume in this section that (A, V) are real valued]. The image of this map $U(A, V)$ is a subgroup of U and it is called the monodromy group of the connection $\mathcal{A} = \sum_{j=1}^n A_j dx_j - V dt$ (cf. Ref. 20). We will often write $\mathcal{A}$ instead of (A, V) . Therefore there is a one-to-one correspondence between the gauge equivalence classes of electromagnetic potentials and the monodromy groups. It was shown by Aharonov and Bohm (cf. Ref. 1) that potentials belonging to different gauge equivalence classes produce a different physical impact and it can be detected in experiments. This phenomenon is called the Aharonov–Bohm effect.

The Aharonov–Bohm assertion is twofold.

- (1) Different gauge equivalent classes of potentials have a different physical impact.
- (2) This impact can be detected in the physical experiment.

Aharonov and Bohm provided such experiment in the case of the magnetic field shielded inside one obstacle.

The solution of the inverse problem proves that any two electromagnetic potentials belonging to different gauge equivalence classes will have different boundary data and therefore are detectable by experiments.

Let S be a two-dimensional smooth surface in D such that $\partial S = \gamma$. It is convenient to represent the electromagnetic field (E, B) as a two-form $\mathcal{F} = \sum_{1 \leq j < k \leq n} B_{jk} dx_j \wedge dx_k + \sum_{k=1}^n E_k dx_k \wedge dt$, where $B_{jk} = -B_{kj}$. Note that $\mathcal{F} = d\mathcal{A}$. Therefore $B_{kj} = \partial A_j / \partial x_k - \partial A_k / \partial x_j$, $E_j = -\partial A_j / \partial t - \partial V / \partial x_j$, $1 \leq j \leq n$, $1 \leq k \leq n$. When $n=3$ we have $B = (B_1, B_2, B_3)$, $E = (E_1, E_2, E_3)$, $B_1 = B_{23}$, $B_2 = -B_{13}$, $B_3 = B_{12}$. Therefore, when $n=3$

$$B = \text{curl } A, \quad E = -\frac{\partial A}{\partial t} - \frac{\partial V}{\partial x}. \quad (4.4)$$

Using the Stoke's formula $\int_{\gamma} \mathcal{A} = \int_S d\mathcal{A}$ we get (cf. Ref. 17)

$$\int_{\gamma} A \cdot dx - V dt = \int_S (B_1 dx_2 dx_3 + B_2 dx_1 dx_3 + B_3 dx_1 dx_2 + E_1 dt dx_1 + E_2 dt dx_2 + E_3 dt dx_3). \quad (4.5)$$

The right hand side of (4.5) is called the electromagnetic flux. Note that the electromagnetic flux does not depend on the surface S such that $\partial S = \gamma$. This follows from the three-dimensional Stoke's theorem and the equality $d\mathcal{F} = d(d\mathcal{A}) = 0$.

In the case $n=2$ formula (4.5) for the electromagnetic flux has the form

$$\int_{\gamma} A_1 dx_1 + A_2 dx_2 - V dt = \int_S (B_3 dx_1 dx_2 + E_1 dt dx_1 + E_2 dt dx_2). \quad (4.6)$$

Formulas (4.5) and (4.6) can be used for the computations of $R(A, V, \gamma)$, especially when either the magnetic or electric fields is “shielded” inside the obstacles, i.e., when either B or E is nonzero in $\Omega'(t) = \bigcup_{j=1}^m \Omega_j(t)$ and zero in D .

The following examples give a description of gauge equivalence classes of electromagnetic potentials using the electromagnetic fluxes.

Consider first an example when electromagnetic field is time independent. Let $n=3$, $D_0 = \{x: |x| < r\}$ and the single obstacle D_1 is a torus inside D_0 as in Remark 3.3 in Ref. 6. Let $B(x)$ be a magnetic field in D_0 such that $\text{supp } B \subset D_1$, i.e., B is shielded in D_1 . Denote by b_0 the flux of B over arbitrary cross section S_0 of the torus D_1 .

Note that b_0 does not depend on the choice of the cross section since $\text{div } B = 0$. Let A be the magnetic potential in D_0 , i.e., $\text{curl } A = B$ in D_0 . Since $B = 0$ in $D_0 \setminus D_1$ the integral $\int_{\gamma} A \cdot dx$ depends only on the homotopy class of γ in $D_0 \setminus D_1$. If B' is another magnetic field in D_0 shielded in D_1 and $\text{curl } A' = B'$ in D_0 then potentials (A, V) and (A', V') are gauge equivalent in $D_0 \setminus D_1$ if and only if $V' = V$ and there exists $m_0 \in \mathbf{Z}$ such that the flux $\int_{S_0} (B' \cdot n) ds = b_0 + 2\pi m_0$, where n is the unit vector normal to S_0 .

In the case of time-dependent electromagnetic potential the situation is more complicated because both the magnetic and the electric field cannot be shielded inside the obstacle. Moreover, we have to use the flux (4.5) instead of the magnetic flux only.

Consider the case $n=2$, $\Omega_0 = \{x: |x| < r\}$ with an obstacle Ω_1 moving with speed v_0 in the direction of x_1 -axis, $\Omega_1 = \bigcup_{t \in (0, T)} \Omega_1(t)$, where $\Omega_1(t) = \{(x, t): (x_1 - v_0 t)^2 + x_2^2 \leq r_1^2\}$, $t \in [0, T]$, $r_1 < r$ is small. Here $D = (\Omega_0 \times (0, T)) \setminus \Omega_1$. In this case $G(D)$ consists of the functions $c(x, t)$ having the form

$$c(x, t) = e^{im\theta + i\psi(x, t)}, \quad (4.7)$$

where $m \in \mathbf{Z}$, $\psi(x, t) \in C^\infty(\bar{D})$, and θ is the angle in polar coordinates centered at $(v_0 t, 0)$ with $x_2 = 0$, $x_1 - v_0 t > 0$ as the polar axis. Consider an electric field E shielded by the “moving” obstacle Ω_1 . Suppose $E = (0, E_2)$ where

$$E_2(x, t) = \delta(x_1 - v_0 t) \delta(x_2) e(t), \quad (4.8)$$

$e(t) \in C^\infty$, $e(t) = \text{const}$ when $t \notin [t_1, t_1 + \varepsilon]$, where $t_1 > 0$, $t_1 + \varepsilon < T$. Let $\Omega_1(x_{10})$ be the intersection of the plane $x_1 = x_{10}$ with Ω_1 and let $\gamma(x_{10})$ be a simple close curve in $(\Omega_0 \times (0, T)) \cap \{x_1 = x_{10}\}$ encircling $\Omega_1(x_{10})$. By formula (4.6) we have

$$\int_{\gamma(x_{10})} A_2 dx_2 - V dt = \int_{\Omega_1(x_{10})} E_2 dx_2 dt = \frac{1}{v_0} e\left(\frac{x_{10}}{v_0}\right). \quad (4.9)$$

The right hand side of (4.9) is the electric flux over $\Omega_1(x_{10})$ and $V(x, t)$ is an electric potential corresponding to $E = (0, E_2)$. Note that (4.9) holds with $e(t)$ replaced by 0 for any closed Jordan curve in $\{x_1 = x_{10}\} \setminus \Omega_1$. Let $B_3(x, t)$ be the magnetic field caused by the electric field $(0, E_2)$. It follows from Maxwell's equations that

$$\frac{\partial E_2}{\partial x_1} = \frac{\partial B_3}{\partial t}.$$

We get from (4.8)

$$B_3(x, t) = -\frac{1}{v_0} \delta(x_1 - v_0 t) \delta(x_2) e(t) - \frac{1}{v_0^2} \theta(x_1 - v_0 t) \delta(x_2) \frac{\partial e\left(\frac{x_1}{v_0}\right)}{\partial t}, \quad (4.10)$$

where $\theta(\tau) = 1$ when $\tau > 0$, $\theta(\tau) = 0$ when $\tau < 0$.

Let $\gamma(t)$ be a closed simple curve in $(\Omega_0 \cap \{t = \text{const}\}) \setminus \Omega_1(t)$. We have by (4.6)

$$\int_{\gamma(t)} A \cdot dx = \int_{S(t)} B_3 dx_1 dx_2 = -\frac{1}{v_0} e(t) - \frac{1}{v_0^2} \int_{S(t)} \theta_1(x_1 - v_0 t) \delta(x_2) \frac{\partial e\left(\frac{x_1}{v_0}\right)}{\partial t} dx_1 dx_2,$$

where $\partial S(t) = \gamma(t)$. Since $\partial/\partial t e(x_1/v_0) = v_0(\partial/\partial x_1)e(x_1/v_0)$ we get

$$\int_{S(t)} \theta(x_1 - v_0 t) \delta(x_2) \frac{\partial e\left(\frac{x_1}{v_0}\right)}{\partial t} dx_1 dx_2 = v_0 \left(e\left(\frac{\tilde{x}_1(t)}{v_0}\right) - e(t) \right),$$

where $\tilde{x}_1(t)$ is the intersection of the line $x_1 = v_0 t$, $x_2 = 0$ with $\gamma(t)$. Therefore

$$\int_{S(t)} B_3 dx_1 dx_2 = -\frac{1}{v_0} e\left(\frac{\tilde{x}_1(t)}{v_0}\right). \quad (4.11)$$

Note that knowing the electric flux (4.9) for all x_{10} we can determine the magnetic flux (4.11). Let $E'_2(x, t) = \delta(x_1 - v_0 t) \delta(x_2) e'(t)$ be shielded by the obstacle Ω_1 , where $e'(t) = \text{const}$ when $t \in [t_1, t_1 + \varepsilon]$, and let B'_3 be the corresponding magnetic field. Let (A, V) and (A', V') be any corresponding electromagnetic potentials in $\Omega_0 \times (0, T)$. If (A, V) and (A', V') are gauge equivalent in D then there exists $m \in \mathbb{Z}$ such that

$$\frac{1}{v_0} e'\left(\frac{x_1}{v_0}\right) = \int_{\Omega_1(x_1)} E'_2 dx_2 dt = \int_{\Omega_1(x_1)} E_2 dx_2 dt + 2\pi m = \frac{1}{v_0} e\left(\frac{x_1}{v_0}\right) + 2\pi m \quad (4.12)$$

for all x_1 . Vice versa, suppose (4.12) holds. Then for the magnetic fluxes we also have

$$\int_{S(t)} B'_3 dx_1 dx_2 = \int_{S(t)} B_3 dx_1 dx_2 + 2\pi m$$

for any $S(t)$ such that $\gamma(t) = \partial S(t)$ is simple closed curve in $\Omega_0 \setminus \Omega_1(t)$. Moreover, (4.10) and (4.12) imply that $B_3 - B'_3 = (1/v_0) \delta(x_1 - v_0 t) \delta(x_2) 2\pi m = 0$ in $\Omega_0 \setminus \Omega_1(t)$ for all t . This means that $B_3 - B'_3$ as $E_2 - E'_2$ are confined to the obstacle Ω_1 . Therefore for any close path γ in $(\Omega_0 \times (0, T)) \setminus \Omega_1$ the line integral $\int_{\gamma} (A' - A) \cdot dx + (V - V') dt$ depends only on the homotopy class of γ . Making a homotopy of γ to a multiple of the $\gamma(x_{10})$ we get, using (4.12), that $\int_{\gamma} (A' - A) \cdot dx + (V - V') dt = 2\pi m'$, where $m' \in \mathbb{Z}$. Therefore (A, V) and (A', V') are gauge equivalent.

Finally, let $n=2$ and Ω_0, Ω_1 be the same as above. Consider an example of the electromagnetic fluxes caused by the magnetic field $B_3(x, t) = \delta(x_1 - v_0 t) \delta(x_2) b(t)$ in $\Omega_0 \times (0, T)$ shielded in Ω_1 . In this case we can allow $v_0=0$, i.e., the obstacle Ω_1 may not move. Let $\gamma(t_0)$ be a closed simple curve in $\Omega_0 \times \{t=t_0\}$ that encircles $\Omega_1(t_0)$. Then by (4.6)

$$\int_{\gamma(t_0)} A \cdot dx = \int_{\Omega_1(t_0)} B_3 dx_1 dx_2 = b(t_0).$$

Here $A(x, t)$ are arbitrary magnetic potentials in $\Omega_0 \times (0, T)$ such that $\text{curl } A = B(x, t)$. For example, we may take

$$A = (A_1(x), A_2(x)) = \frac{b(t)}{2\pi} \left(\frac{-x_2}{(x_1 - v_0 t)^2 + x_2^2}, \frac{x_1 - v_0 t}{(x_1 - v_0 t)^2 + x_2^2} \right).$$

Let $B'_3(x, t)$ be an arbitrary magnetic field shielded in Ω_1 and such that

$$\int_{\Omega_1(t)} B'_3(x, t) dx_1 dx_2 = b(t) + 2\pi m \quad (4.13)$$

for some $m \in \mathbf{Z}$. Denote by A' an arbitrary magnetic potential in $\Omega_0 \times (0, T)$ such that $\text{curl } A' = B'_3$ in $\Omega_0 \times (0, T)$. Then we have

$$\exp \left(i \int_{\gamma(t)} A \cdot dx \right) = \exp \left(i \int_{\gamma(t_0)} A' \cdot dx \right)$$

for any closed path in $(\Omega_0 \times \{t=t_0\}) \setminus \Omega_1(t_0)$, $\forall t_0 \in [0, T]$. Therefore there exists $c(x, t) \in G(D)$ such that $A' - A = ic^{-1}(\partial c / \partial x)$ in D .

Define $V' = V - ic^{-1}(\partial c / \partial t)$. Then (A', V') is gauge equivalent to (A, V) . Vice versa, if (A', V') and (A, V) are gauge equivalent then (4.13) holds.

Note that in this example the magnetic field is shielded in Ω_1 but the electric field $E = -\partial V / \partial x - \partial A / \partial t$ is not shielded in Ω_1 . In the previous example the electric field E was shielded in Ω_1 but the magnetic field was not.

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